INHA UNIVERSITY TASHKENT
DEPARTMENT OF CSE & ICE

Fall 2023 – Application Programming in Java
Lab Assignment # 5

Topic covered Arithmetic Operator and I/O in Java
Last Date of Submission: 12/11/2023

INSTRUCTIONS:

1. Program should be well commented.
2. Program should be easy to use.
3. All Lab assignments are to be completed by an individual student on his/her computer system and not in groups

Problems :

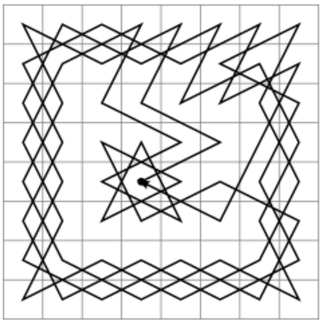
INSTRUCTIONS: For this *assignment*, you are required to develop a **Java Graphical User Interface (GUI) Application** to demonstrate you can use *Java* constructs including input/output via a GUI, *Java* primitive and built-in types, *Java* defined objects, arrays, selection and looping statements and various other *Java* commands. Your program must produce the correct results.

P.1 Knight's tour (50 points)

Create **Java Graphical User Interface (GUI) Application** to demonstrate solution for following task:

A knight's tour is a sequence of moves of a knight on a chessboard such that the knight visits every square only once. If the knight ends on a square that is one knight's move from the beginning square (so that it could tour the board again immediately, following the same path), the tour is closed, otherwise it is open.

Example



Task: Implement knight's tour algorithm and demonstrate it to user using GUI.

1. You need to ask user to enter the dimension n for chessboard (n x n).
2. Ask user to enter the initial position for knight.
3. Generate correct sequence of moves for knight for this board
4. Draw a board (n x n).
5. Show knight's tour to user.

Key points:

1. After each move knight should wait while user press space or button on form.
2. Knight should leave trace and mark each visited cell.

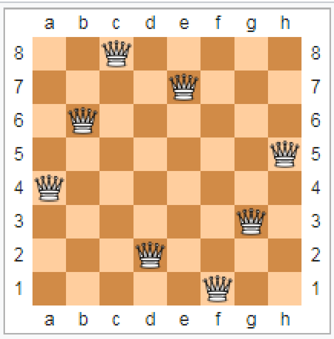
P2. N queens puzzle (50 points)

The **N queens puzzle** is the problem of placing N chess queens on an N×N chessboard so that **no two queens threaten each other**. Thus, a solution requires that no two queens share the same row, column, or diagonal. Solutions for this problem exist for all natural numbers *n* with the exception of *n*=2 and *n*=3.

Task: Implement N queens puzzle algorithm and demonstrate it to user using GUI.

1. You need to ask user to enter the dimension n for chessboard (n x n).
2. Find right positions for n queens for this board
3. Draw a board (n x n).
4. Show queens to user.

Examples:



Explicit solution for 8 queens

Explicit solution for 9 queens

Explicit solution for 10 queens

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